Which Loans Are Worth Funding?

Credit risk and lending profitability analysis | Tejas Sharma | Python, SQL, logistic regression

The question

Every lender has to decide who to lend to and whether the interest earned from borrowers who repay covers the losses from those who don't. This project answers that question using about 164,000 completed personal loans from Lending Club, a US peer-to-peer lender, with each borrower's details at application and whether the loan went bad.

Approach

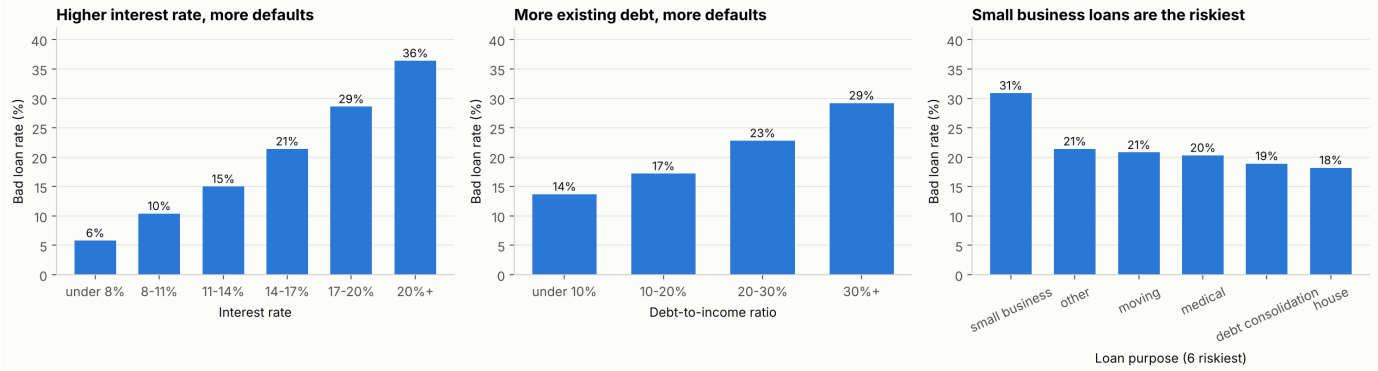
- **Explored the data with SQL** to find which borrower and loan features go with higher default rates.
- **Built a logistic regression model** to estimate each loan's probability of default (PD), trained on 70% of loans and tested on the 30% it never saw.
- **Converted each PD into expected profit**, using the interest rate, a 4% cost of funds and a 60% loss given default.
- **Compared two lending policies** on the unseen loans: fund every loan, or fund only loans with positive expected profit.

Results on 49,139 unseen loans

	Fund every loan	Model policy
Loans funded	49,139	38,998 (79%)
Amount lent	$642.7m	$521.8m
Bad loan rate	18.3%	16.5%
Actual profit	$34.0m	**$37.6m**
Return on amount lent	5.3%	**7.2%**

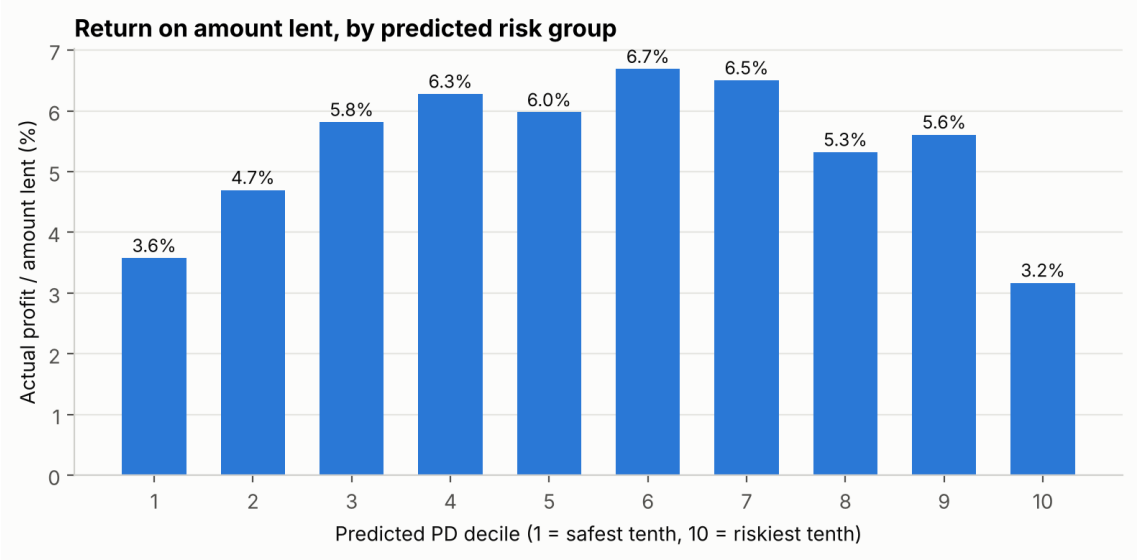
Lending only where expected profit is positive earned **$3.6m (11%) more profit while lending $121m (19%) less**, and cut the bad loan rate from 18.3% to 16.5%.

What drives default



Bad loan rate by interest rate, debt-to-income ratio and loan purpose (all 163,794 loans).

Where the profit is



Actual return on amount lent for each tenth of the unseen loans, ranked by predicted PD.

Key findings

- **Default risk is predictable from application data.** Interest rate, loan size relative to income, debt-to-income, a 60-month term, small business purpose and credit card utilisation raise risk; higher income lowers it. The model reaches an AUC of 0.70 (Gini 0.41), and its predicted default rates match actual rates in every risk group.
- **The riskiest loans were underpriced.** The riskiest tenth defaulted at about 41% and, even at an average rate of about 21%, returned only about 3% on the amount lent, against around 6.5% for mid-risk loans.
- **Lending on expected profit beats lending to everyone.** The gain holds across loss-given-default assumptions and grows as losses get more severe: $0.2m extra profit at 40% LGD, $3.6m at 60% and $14.1m at 80%.
- **A pattern is not a cause.** Borrowers with verified income defaulted more often, but only because riskier applicants were the ones being verified. Once the model accounts for price and finances, verification no longer signals risk.

What a lender would do next

- Re-price the riskiest segments instead of declining them outright, and test whether demand holds at the higher rate.
- Validate the policy on the most recent loans rather than a random split, to check it holds as the economy changes.
- Add credit bureau and bank transaction data to separate good and bad borrowers more sharply.

Limitations

- A single loss given default for all loans and no operating costs.
- Only approved loans are in the data, so declined applicants are never seen.
- No application dates, so the train/test split is random rather than by time.